Mengdi Wang

(+49)15129092305 · wangmengdi0701@gmail.com
wmd0701.github.io · github.com/wmd0701 · Scholar

Education

TU Munich, *Germany, PhD* 2022.08 - 2026.06 (estimated)

- **Focus**: Collaborative ML, eye tracking, XR, LLM in XR
- **Thesis**: Preserving Eye Privacy in XR: Local ML Pipelines From Iris Obfuscation to Collaborative ML

ETH Zurich, *Switzerland, Master in Computer Science* 2019.09 - 2022.07

- **Thesis**: Exploration of Deep Features for Neural Style Transfer

TU Munich, *Germany, Bachelor in Informatics: Games Engineering* 2016.09 - 2019.07

- **Thesis**: Porting MLEM Algorithm for Heterogeneous Systems

Publications

- **Wang, M.**, Bozkir, E., & Kasneci, E. (2026). CycleSL: Server-Client Cyclical Update Driven Scalable Split Learning. *In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 1841-1851.*
- Bozkir, E., Özdel, S., **Wang, M.**, David-John, B., Gao, H., Butler, K., Jain, E., & Kasneci, E. (2026). Eye-Tracked Virtual Reality: A Comprehensive Survey on Methods and Privacy Challenges. *Proceedings of the IEEE (PIEEE).*
- Bodonhelyi, A., **Wang, M.**, Bozkir, E., Bühler, B., & Kasneci, E. (2026). Safeguarding Privacy: Privacy-Preserving Detection of Mind Wandering and Disengagement Using Federated Learning in Online Education. *arXiv preprint arXiv:2602.09904.*
- **Wang, M.**, Bozkir, E., & Kasneci, E. (2025). Iris Style Transfer: Enhancing Iris Recognition with Style Features and Privacy Preservation through Neural Style Transfer. *Proceedings of the ACM on Computer Graphics and Interactive Techniques (PACMCGIT), 8(2), 1-21; the 2025 ACM Symposium on Eye Tracking Research and Applications (ETRA).*
- **Wang, M.**, Bozkir, E., & Kasneci, E. (2025). Trade-Offs in Privacy-Preserving Eye Tracking through Iris Obfuscation: A Benchmarking Study. *In 2025 25th International Conference on Digital Signal Processing (DSP), pp. 1-5.*
- Buldu, K. B., Özdel, S., Lau, K. H. C., **Wang, M.**, Saad, D., Schönborn, S., Boch, A., Kasneci, E., & Bozkir, E. (2025). CUIfy the XR: An Open-Source Package to Embed LLM-Powered Conversational Agents in XR. *In 2025 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR), pp. 192-197.*
- Bozkir, E., Özdel, S., Lau, K. H. C., **Wang, M.**, Gao, H., & Kasneci, E. (2024). Embedding Large Language Models into Extended Reality: Opportunities and Challenges for Inclusion, Engagement, and Privacy. *In Proceedings of the 6th ACM Conference on Conversational User Interfaces (CUI), pp. 1-7.*
- **Wang, M.**, Bodonhelyi, A., Bozkir, E., & Kasneci, E. (2024). TurboSVM-FL: Boosting Federated Learning through SVM Aggregation for Lazy Clients. *In Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), Vol. 38, No. 14, pp. 15546-15554.*

Experience

TU Munich, *Chair HCTL, Research Assistant* 2022.08 - 2026.03

- Demonstrated novel federated and split learning algorithms (TurboSVM-FL and CycleSL) that improve model performance up to **30%** and convergence by **2x** in heterogeneous data distribution and sparse client participation without incurring any extra client computational burden or data transfer overhead

- Developed novel iris recognition method leveraging style features, demonstrating **2%** higher recognition accuracy and superior robustness against image variation in contrast to existing approaches. Accordingly, applied neural style transfer for privacy-preserving iris obfuscation with minimal utility loss and low spoofing risk
- Built real-time LLM-driven interaction system for Pepper robot (STT + TTS + joystick control), solved compatibility issues, improved user experience, and deployed in live interaction sessions with children
- Designed plug-and-play Unity module enabling LLM-based interaction with STT and TTS for XR apps
- Maintained computational server with 4x NVIDIA A100 and 1x H200 GPUs and 20x desktops with RTX 4080
- Reviewed manuscripts for top-tier conferences including ICML, AAAI, NeurIPS, CHI, AIED, ETRA
- Composed research proposal for applying DFG funding following dissertation topic (proposal already granted)

ETH Zurich, *Computer Graphics Lab, Research Student* 2022.02 - 2022.07

- Evaluated various factors influencing the 2D-to-3D stylization process utilizing differentiable renderer
- Presented guidelines for style feature decomposition and controlling stroke granularity, and introduced a novel way for per-vertex color initialization to align reshaping and texturing outcomes

Non-Research Projects

CloudVis, *Unity / C#*

- Simulated large-scale atmospheric processes like clouds and air flows, supporting real-time interaction
- Implemented shaders to support volume and vector field visualization

SplashyWaves, *C++*

- Simulated fluid dynamics in moving container using PBD method, yielding improved stability compared to SPH

Kings of Dominia, *Unity / C#*

- Designed multi-platform touchscreen video game emulating the physics-based domino toppling game

Others

- **Technical Skills**: Python (PyTorch), LM Studio, ComfyUI, Agent CLI, C# (Unity), ROCm, CUDA, Linux, Git
- **Languages**: Chinese (native), English (C1), German (C1), Japanese (beginner)
- **Teaching**: Human-AI Interaction, Serious Games in Extended Reality, Recent Advances in Privacy, Introduction to Programming and Data Processing